Lecture 10

Differential forms

Objectives. *A differential form assigns an alternating multilinear measurement to tangent vectors at each point. Forms pull back under every smooth map. Their exterior derivative extends the differential of a function and combines the Cartesian gradient, curl and divergence. We construct this derivative in coordinates, prove that it is independent of coordinates, and compute the sphere's area form.*

By the end of this lecture you will be able to:

- Evaluate forms with the determinant wedge convention.
- Compute exterior derivatives and pullbacks under smooth maps.
- Detect nonexactness of a closed 1-form by a nonzero loop integral.
- Derive both angular and stereographic expressions for sphere area.

10.1 Covectors and the cotangent bundle

Let M be a smooth n-manifold. The *cotangent space* at $p \in M$ is the dual vector space $T_p^*M = \mathrm{Hom}(T_pM, \mathbb{R})$. Its elements are *covectors*. For a smooth function $f\colon M \to \mathbb{R}$, its *differential* is given by

$$(\mathrm{d}f)_p(v) = v(f) \qquad (v \in T_pM).$$

This is the differential of Lecture 8 with $T_{f(p)}\mathbb{R}$ identified with $\mathbb{R}$ by its standard coordinate. In coordinates $x^1, \dots, x^n$, the covectors $\mathrm{d}x^i$ are dual to ∂_j, since $\mathrm{d}x^i(\partial_j) = \delta^i_j$. Thus a covector has a unique expression $\alpha = \alpha_i \mathrm{d}x^i$, and $\alpha(v) = \alpha_i v^i$. In particular,

$$\mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i.$$

The *cotangent bundle* is the disjoint union $T^*M = \coprod_{p\in M} T_p^*M$, with projection to p. Its local coordinates are $(x^1, \dots, x^n, \alpha_1, \dots, \alpha_n)$. If $y = y(x)$ has Jacobian $J = Dy$, invariance of $\alpha(v)$ requires the coefficient row to change by $\alpha' = \alpha J^{-1}$. These are smooth invertible bundle transitions. They give the cotangent manifold by the same construction as sketched for the tangent bundle in Lecture 8. The Hausdorff and countable-basis arguments there apply with these transitions.

A *smooth 1-form* is a smooth section of T^*M. Equivalently, its coefficients α_i are smooth in every chart. The form $\mathrm{d}f$ is smooth because the partial derivatives of f are smooth. A covector measures a tangent vector, as the figure illustrates.

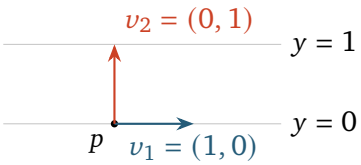


Figure 10.1: The horizontal lines are levels of the coordinate y. The covector $\mathrm{d}y$ gives 0 on the horizontal vector and 1 on the vertical vector.

For example, on $\mathbb{R}^2$ let $\alpha = 2x\mathrm{d}x + \mathrm{d}y$. At $(1, 0)$ it gives $\alpha(3\partial_x - \partial_y) = 6 - 1 = 5$. This form is the differential of $x^2 + y$.

10.2 Alternating tensors and the wedge product

A *tensor field of type* (r, s) assigns at each point a multilinear real-valued map with r cotangent arguments and s tangent arguments. Its coordinate coefficients must be smooth. A tensor with $r = 0$ is *covariant*. A *k-form* is a covariant k-tensor field that is alternating: interchanging two arguments reverses its sign. We write $\Omega^k(M)$ for the vector space of smooth k-forms. A 0-form is a smooth function. Alternation implies that every k-form is zero if $k > \dim M$.

Recall the exterior powers from *Linear Algebra*, Lecture 12, the wedge-and-determinant example.

Proposition 10.1 (Exterior powers · *imported definition; evaluation proved*)**.** *For a finite-dimensional real vector space V, the space $\Lambda^k V^*$ consists of alternating k-linear forms on V. For covectors $\alpha_1, \dots, \alpha_k$, the determinant convention gives*

$$(\alpha_1 \wedge \cdots \wedge \alpha_k)(v_1, \dots, v_k) = \det\big(\alpha_i(v_j)\big)_{i,j=1}^{k}.$$

Proof of the evaluation formula. LA12 identifies covariant tensors with multilinear maps. Its antisymmetric tensors are therefore alternating maps. Evaluating its unnormalized antisymmetrization gives the sum $\sum_\pi \operatorname{sgn}(\pi) \prod_j \alpha_{\pi(j)}(v_j)$. The Leibniz determinant formula identifies this with the stated determinant. ∎

For an increasing index tuple $I = (i_1 < \cdots < i_k)$, write $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}$. Every k-form has a unique expansion

$$\omega = \sum_I a_I \mathrm{d}x^I, \qquad a_I = \omega(\partial_{i_1}, \dots, \partial_{i_k}).$$

To prove this, the determinant gives $\mathrm{d}x^I(\partial_{j_1}, \dots, \partial_{j_k}) = \delta^I_J$ for increasing tuples I and J. This forces the displayed coefficients. Conversely, subtracting the displayed sum from ω gives an alternating form vanishing on every increasing tuple of coordinate vectors. Alternation makes it zero on every ordered tuple, including repeated indices. Multilinearity then makes it zero on arbitrary vectors. This proves existence and uniqueness of the expansion. The sum runs over increasing tuples only, so it has no factorial. If instead every index ranges independently from 1 to n, set $\omega_{i_1 \cdots i_k} = \omega(\partial_{i_1}, \dots, \partial_{i_k})$. Then

$$\omega = \frac{1}{k!} \sum_{i_1, \dots, i_k} \omega_{i_1 \cdots i_k} \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}.$$

Each increasing tuple occurs $k!$ times with compensating signs in the component and basis wedge, which explains the factorial.

The *wedge product* of forms of degrees k and ℓ is the bilinear extension of concatenation of these basis wedges. A repeated index gives zero; otherwise sorting the concatenated indices gives its permutation sign. This definition agrees with the determinant convention. Intrinsically it is

$$(\alpha \wedge \beta)(v_1, \dots, v_{k+\ell}) = \frac{1}{k!\ell!} \sum_{\pi \in S_{k+\ell}} \operatorname{sgn}(\pi) \alpha(v_{\pi(1)}, \dots, v_{\pi(k)}) \beta(v_{\pi(k+1)}, \dots, v_{\pi(k+\ell)}).$$

Indeed, grouping the permutations by which k arguments enter α leaves $k!\ell!$ identical contributions within each group. By bilinearity take $\alpha = \mathrm{d}x^I$ and $\beta = \mathrm{d}x^J$. Both the permutation expression and the sorted basis wedge are alternating in their arguments, so the preceding basis test reduces their comparison to increasing tuples of coordinate vectors. Only shuffles placing the first index block in its matching slots contribute. They give its sorting sign; a repeated index gives zero. Thus the formula equals the sorting rule. The intrinsic formula proves that wedge is independent of coordinates and has smooth coefficients. Concatenating three index lists proves associativity: both parenthesizations have the same sorting sign, or both vanish because an index repeats. Moving a block of k indices past a block of ℓ indices makes $k\ell$ swaps. Bilinearity therefore gives graded commutativity:

$$\alpha \wedge \beta = (-1)^{k\ell} \beta \wedge \alpha.$$

For example, $(\mathrm{d}x + 2\mathrm{d}y) \wedge (3\mathrm{d}x - \mathrm{d}y) = -7\mathrm{d}x \wedge \mathrm{d}y$. Evaluating this on $(2, 1)$ and $(1, -1)$ gives 21. For variable coefficients the same rule gives

$$(x\mathrm{d}x + y\mathrm{d}y) \wedge (z\mathrm{d}x + x\mathrm{d}y) = (x^2 - yz)\mathrm{d}x \wedge \mathrm{d}y.$$

For two orthonormal one-particle states, *QM-I*, Lecture 14, writes the normalized antisymmetric tensor as $(u \otimes v - v \otimes u)/\sqrt{2}$. That state convention includes a norm factor absent from the determinant wedge on covectors.[1] As a tensor, our two-factor wedge is $\alpha \otimes \beta - \beta \otimes \alpha$. For k factors the normalized state uses $1/\sqrt{k!}$ times LA12's unnormalized antisymmetrization. This states the tensor-factor translation without identifying states with covectors. Here $(\mathrm{d}x \wedge \mathrm{d}y)(\partial_x, \partial_y) = 1$. This value also fixes the normalization used by the Hodge operator in Lecture 15.

10.3 The exterior derivative

The *exterior derivative* raises form degree by one. On a coordinate chart define it by

$$\mathrm{d}\omega = \sum_I \sum_j (\partial_j a_I) \mathrm{d}x^j \wedge \mathrm{d}x^I \quad \text{when} \quad \omega = \sum_I a_I \mathrm{d}x^I.$$

On functions this agrees with the differential already defined.

Theorem 10.2 (Exterior derivative · *sketched; existence and identities proved*)**.** *The coordinate formula defines a coordinate-independent real-linear map* $\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M)$*. It satisfies* $\mathrm{d}^2 = 0$ *and, for* $\alpha \in \Omega^k(M)$*,*

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^k \alpha \wedge \mathrm{d}\beta.$$

These properties and its value on functions determine it uniquely.

[1] The source is QM-I, Lecture 14, Many fermions: Slater determinants. Its tensor normalization does not change the present wedge.

Proof of existence and identities. Work first in one chart. The formula $\mathrm{d}(b\mathrm{d}x^J) = \mathrm{d}b \wedge \mathrm{d}x^J$ also holds for a tuple J that is not increasing: reordering changes both sides by the same sign, and a repeated index makes both sides zero. Applying d twice gives terms $(\partial_r \partial_j a_I)\mathrm{d}x^r \wedge \mathrm{d}x^j \wedge \mathrm{d}x^I$. Terms with $r = j$ vanish. Each remaining term cancels its partner with j and r exchanged, because mixed derivatives agree and wedge is antisymmetric. Thus $\mathrm{d}^2 = 0$.

For the product rule, expand into terms $\alpha = a\mathrm{d}x^I$ and $\beta = b\mathrm{d}x^J$, where $|I| = k$. The ordinary product rule differentiates ab into $b\mathrm{d}a + a\mathrm{d}b$. The first contribution is $\mathrm{d}\alpha \wedge \beta$. In the second, moving $\mathrm{d}b$ past the k factors of $\mathrm{d}x^I$ gives $(-1)^k \alpha \wedge \mathrm{d}\beta$. Summing proves the identity.

Now let $y = y(x)$ be another chart. The chain rule gives $\mathrm{d}y^a = (\partial_i y^a)\mathrm{d}x^i$. The operator defined in x coordinates annihilates $\mathrm{d}y^a$ by the same mixed-derivative cancellation. For a coefficient function b, the chain rule also identifies its x and y differentials. Apply the product rule just proved to a monomial $b\mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_k}$. Only the differentiated coefficient survives, so the result is precisely the y coordinate formula. Such monomials span every form. Thus the definitions agree on chart overlaps and give a global smooth form. ∎

Sketch of uniqueness. Let D have the stated properties. A smooth *cutoff function* equals 1 near a chosen point and 0 outside a chosen chart neighbourhood. Such functions are constructed in a smaller coordinate ball using $e^{-1/t}$ for $t > 0$, extended by zero for $t \le 0$. If α vanishes near p, choose a cutoff χ supported where $\alpha = 0$. Since $\chi\alpha = 0$, the product rule at p gives $(D\alpha)_p = 0$. Thus D depends only on the form near p. Use a cutoff to extend local coefficient and coordinate functions smoothly to M, without changing them near p. The identity $D\mathrm{d}f = D^2 f = 0$ for these extensions and the product rule force $D(\sum_I a_I \mathrm{d}x^I) = \sum_I \mathrm{d}a_I \wedge \mathrm{d}x^I$ at p. This proves uniqueness, with the cutoff construction left at sketch level. See Tu, §19.4. ∎

For example, $\mathrm{d}(x\mathrm{d}y) = \mathrm{d}x \wedge \mathrm{d}y$, while $\mathrm{d}(x\mathrm{d}x) = \mathrm{d}x \wedge \mathrm{d}x = 0$.

10.4 Pullback under a smooth map

Let $f\colon M \to N$ be any smooth map. For $\omega \in \Omega^k(N)$ its *pullback* is given by

$$(f^*\omega)_p(v_1, \ldots, v_k) = \omega_{f(p)}(\mathrm{d}f_p v_1, \ldots, \mathrm{d}f_p v_k).$$

For functions, set $f^*h = h \circ f$. The differential sends each argument to the required target tangent space. No inverse of f or of its differential is needed.

Theorem 10.3 (Pullback calculus). *Pullback takes smooth forms to smooth forms and satisfies*

$$f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta, \qquad \mathrm{d}(f^*\alpha) = f^*(\mathrm{d}\alpha).$$

For smooth $g\colon N \to P$ it also satisfies $(g \circ f)^* = f^*g^*$ and $\mathrm{id}^* = \mathrm{id}$.

Proof. Substituting $\mathrm{d}f_p v_j$ in the permutation formula proves wedge compatibility. The chain rule gives $f^*\mathrm{d}h = \mathrm{d}(h \circ f)$. In local target coordinates y, write $f^a = y^a \circ f$. The pullback of $\alpha = \sum_I a_I \mathrm{d}y^I$ is

$$f^*\alpha = \sum_I (a_I \circ f)\mathrm{d}f^{i_1} \wedge \cdots \wedge \mathrm{d}f^{i_k}.$$

Its coefficients are smooth. Differentiating this expression leaves only $\mathrm{d}(a_I \circ f)$, since $\mathrm{d}^2 f^a = 0$. Pulling back $\mathrm{d}\alpha$ gives the same expression by the chain rule on coefficient functions. This proves commutation with d without using the uniqueness sketch. Finally, substituting $\mathrm{d}(g \circ f) = \mathrm{d}g \circ \mathrm{d}f$ into the definition proves the composition rule, and identity differentials prove the identity rule. ∎

For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, pullback gives $f^*(\mathrm{d}y) = 2x\mathrm{d}x$, including at $x = 0$. Lecture 9 showed that ∂_x does not descend under this same map. Forms pull back even when a map loses rank or identifies distinct points.

10.5 Gradient, curl and divergence

Fix Cartesian coordinates on Euclidean $\mathbb{R}^3$. Identify a vector (A_1, A_2, A_3) with the 1-form $\alpha = A_1\mathrm{d}x + A_2\mathrm{d}y + A_3\mathrm{d}z$. Identify (B_1, B_2, B_3) with the 2-form $\beta = B_1\mathrm{d}y\wedge\mathrm{d}z + B_2\mathrm{d}z\wedge\mathrm{d}x + B_3\mathrm{d}x\wedge\mathrm{d}y$. Identify a scalar g with the 3-form $g\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$. These are specified coordinate identifications.

Proposition 10.4 (Vector calculus from forms). *Under these identifications, exterior differentiation gives gradient, curl and divergence in successive degrees.*

Proof. The defining formula gives the following coefficients:

$$\begin{aligned}
\mathrm{d}f &= (\partial_x f)\mathrm{d}x + (\partial_y f)\mathrm{d}y + (\partial_z f)\mathrm{d}z,\\
\mathrm{d}\alpha &= (\partial_y A_3 - \partial_z A_2)\mathrm{d}y\wedge\mathrm{d}z + (\partial_z A_1 - \partial_x A_3)\mathrm{d}z\wedge\mathrm{d}x\\
&\quad + (\partial_x A_2 - \partial_y A_1)\mathrm{d}x\wedge\mathrm{d}y,\\
\mathrm{d}\beta &= (\partial_x B_1 + \partial_y B_2 + \partial_z B_3)\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z.
\end{aligned}$$

For example, the $\mathrm{d}y\wedge\mathrm{d}z$ coefficient receives $\partial_y A_3$ from $\mathrm{d}(A_3\mathrm{d}z)$ and $-\partial_z A_2$ from $\mathrm{d}(A_2\mathrm{d}y)$. Cyclic permutation gives the other two coefficients. For $\mathrm{d}\beta$, repeated factors vanish and each surviving cyclic order has positive sign. The displayed coefficients are the Cartesian definitions of gradient, curl and divergence. ∎

Applying $\mathrm{d}^2 = 0$ in degrees zero and one respectively gives

$$\operatorname{curl}\operatorname{grad} f = 0, \qquad \operatorname{div}\operatorname{curl} A = 0.$$

10.6 Closed and exact forms

A form ω is *closed* if $\mathrm{d}\omega = 0$. A k-form with $k \geq 1$ is *exact* if it equals $\mathrm{d}\eta$ for some $(k-1)$-form η. Every exact form is closed by $\mathrm{d}^2 = 0$. For a smooth 1-form α on M and a smooth path $\gamma\colon [a,b] \to M$, define the line integral by

$$\int_\gamma \alpha = \int_a^b \alpha_{\gamma(t)}(\gamma'(t))\,\mathrm{d}t.$$

Here a smooth path on $[a,b]$ means the restriction of a smooth curve on an open interval containing $[a,b]$. The pullback definition gives $\gamma^*\alpha = \alpha_{\gamma(t)}(\gamma'(t))\mathrm{d}t$, so this is also the integral of the pullback's coefficient. If $\alpha = \mathrm{d}f$, the chain rule and the fundamental theorem of calculus give $\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a))$.

Proposition 10.5 (A closed form that is not exact). *On $\mathbb{R}^2 \setminus \{0\}$ the angular form*

$$\omega_{\mathrm{ang}} = \frac{x\mathrm{d}y - y\mathrm{d}x}{x^2 + y^2}$$

is closed but not exact.

Proof. Its exterior derivative has $dx \wedge dy$ coefficient

$$\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2} = \frac{2}{x^2+y^2} - \frac{2(x^2+y^2)}{(x^2+y^2)^2} = 0.$$

For $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, substitution gives $\gamma^* \omega_{\text{ang}} = dt$. Its integral is 2π. If it were exact, the endpoint formula would give zero on this closed path. This contradiction proves nonexactness. ∎

For a smooth loop $\gamma(t) = r(t)(\cos\phi(t), \sin\phi(t))$, where $r(t) > 0$, differentiation gives $x\dot{y} - y\dot{x} = r^2\dot{\phi}$. Thus $\gamma^* \omega_{\text{ang}} = \dot{\phi} dt$. Lecture 3 lifts $\gamma/|\gamma|$ in turns; multiplying that lift by 2π gives ϕ. The angle ϕ is smooth because it locally differs from a smooth angle branch along γ by a constant multiple of 2π. The integral divided by 2π is therefore the winding number. The same computation with variable r and ϕ gives $\omega_{\text{ang}} = d\phi$ on an angle chart. There is no global smooth angle function on the punctured plane.

10.7 The area form of the sphere

The inclusion $i\colon S^2 \to \mathbb{R}^3$ pulls the ambient form $\beta = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy$ back to $\omega_{S^2} = i^*\beta$. At $p \in S^2$, for $v, w \in T_p S^2$, it evaluates as

$$(\omega_{S^2})_p(v, w) = \det[p\ v\ w] = p \cdot (v \times w).$$

This is signed area measured with the outward unit normal p. The next figure shows the sign test at the south pole for the north-pole stereographic chart of Lecture 7.

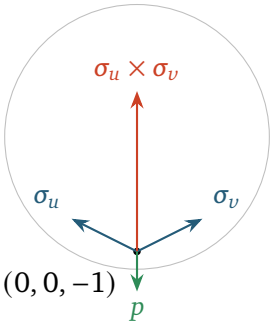


Figure 10.2: Viewed from below the equator, the north chart's tangents at the south pole have cross product $+4e_3$. The outward normal is $-e_3$.

Example 10.6 (Angular and stereographic area)**.** For angular parameters put $s(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, with $0 < \theta < \pi$ and ϕ in an open interval of length less than 2π. Its coordinate tangents are

$$s_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \qquad s_\phi = (-\sin\theta\sin\phi, \sin\theta\cos\phi, 0).$$

Their cross product is $s_\theta \times s_\phi = \sin\theta\, s$, so

$$s^* \omega_{S^2} = \sin\theta\, d\theta \wedge d\phi.$$

For the inverse stereographic map from the north pole, set $D = 1 + u^2 + v^2$ and $\sigma(u, v) = (2u, 2v, u^2 + v^2 - 1)/D$. Its derivative matrix is

$$(\sigma_u\ \sigma_v) = \frac{1}{D^2} \begin{pmatrix} 2(D - 2u^2) & -4uv \\ -4uv & 2(D - 2v^2) \\ 4u & 4v \end{pmatrix}.$$

The cross product of these columns is

$$\sigma_u \times \sigma_v = \frac{(-8uD, -8vD, 4D(1-u^2-v^2))}{D^4} = -\frac{4\sigma}{D^2}.$$

Since $\sigma \cdot \sigma = 1$, the pullback is

$$\sigma^* \omega_{S^2} = -\frac{4}{(1+u^2+v^2)^2} \mathrm{d}u \wedge \mathrm{d}v.$$

At $(u, v) = (0, 0)$ the point is $(0, 0, -1)$, while the tangent columns are $2e_1$ and $2e_2$. Their cross product points opposite to the outward normal. This verifies the negative sign. Lecture 12 formalizes this comparison as orientation reversal.

Self-check. Compute $\mathrm{d}(x\mathrm{d}y \wedge \mathrm{d}z)$ on $\mathbb{R}^3$ and its pullback to S^2.

Answer. The derivative is $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. Its pullback is zero: an alternating trilinear form vanishes on the two-dimensional tangent space of the sphere.

Summary. Forms alternate in their tangent arguments and use the determinant wedge. The exterior derivative is independent of coordinates, squares to zero and satisfies the graded product rule. Pullback exists under every smooth map and commutes with this derivative. A closed form need not be exact, as the angular form's nonzero loop integral proves. The sphere's outward area has negative coefficient in the north-pole stereographic coordinates. Lecture 11 differentiates forms along flows.

Reading. Tu, *An Introduction to Manifolds*, §§17–19; *Linear Algebra*, Lecture 12, exterior powers and the determinant wedge.